

Vivek Ranjan

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Staff Software Engineer with 10+ years track record of leading agile teams and delivering high-impact, scalable, full-stack systems using Python, Java, TypeScript, React, NextJS, and AWS. Enjoys collaborating across teams to solve significant, real-world problems while relentlessly discovering ways to optimize software systems and organizational delivery processes.

SKILLS

BACKEND

Python, NodeJS, Java, SQL (MS SQL Server, Postgres)

FRONTEND

Typescript, React, NextJS

INFRASTRUCTURE/OTHER

Agentic AI Coding (Cursor, Claude Code, Junie, etc), AWS (Lambda, API Gateway, SQS/SNS, ECS, ECR, Fargate, ElasticBeanstalk, Cloudformation, CDK, S3, EC2, RDS, IAM, etc), Datadog, Grafana (Loki, Prometheus, etc), CI/CD pipelines, Docker, Kubernetes (kubectl, Helm, k9s, etc), Agile, Scrum, Shell

WORK EXPERIENCE

Staff Software Engineer

Capsule

April 2025 - Present

- Built multiple products backed by microservices to increase processing speed for pharmacists to validate prescriptions and benefits.
 - Improved database performance leading to 21% reduction in CPU usage, 35% reduction in IOPS and 50% reduction in costs.
 - Backend - Python, pytest, Django, PostgreSQL
 - Frontend - Typescript, NextJS, React, Vite
 - Infrastructure/Other - Docker, Kubernetes, AWS (EKS, RDS, SQS/SNS), Shell, CI/CD pipelines, Sentry, Grafana (Prometheus, Loki)
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Staff Software Engineer

Renaissance Learning

February 2022 - April 2025

- Proposed & led project: Achieved 73% cost reduction by moving batch jobs to AWS Fargate.
 - Created an IaC framework (Python, AWS CDK, Docker) that raised deployment frequency by 40% and streamlined batch-job development for all teams.
 - Mentored multiple engineers leading to their promotions and improvement of their technical skills.
 - Backend - Python, pytest, Java, Spring (MVC), Groovy/Spock, Microsoft SQL Server
 - Frontend - Vue/Vuex, HTML/CSS, JavaScript, Jest
 - Infrastructure/Other - Docker, AWS (ECS/ECR, Fargate, CDK, RDS, Aurora Serverless), Shell, CI/CD pipelines, Datadog, Cronitor
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Lead Software Engineer

Renaissance Learning

June 2019 - February 2022

- Led project: UX redesign implementation of crucial progress monitoring tool used by educators and millions of students. Modernized frontend (Vue, Vuex, Jest) and optimized backend (Java, Spring, REST, Hibernate, MS SQL Server, Groovy/Spock).
- Proposed & led project: Reduced engineering effort by 4 weeks annually by building a microservice driven content-management system on AWS Aurora Serverless, Lambda, and API Gateway with a Vue front-end.

- Proposed & led project: Doubled deployment frequency and saved 10 hrs/week by automating monolith deployments and adding end-to-end CI/CD tests with Docker and CloudFormation
 - Backend - Python, pytest, Java, Spring (MVC), Groovy/Spock, Microsoft SQL Server, AWS Aurora (serverless Postgres)
 - Frontend - Vue/Vuex, HTML/CSS, JavaScript, Jest
 - Infrastructure/Other - Docker, AWS (Lambda, API Gateway, ElasticBeanstalk, CloudFormation, IAM, RDS, Aurora Serverless), Shell, CI/CD pipelines, Datadog
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Founding Engineer

FastBridge Learning

June 2015 - June 2019

- 4th engineer at Seed-stage startup.
- Led project: Delivered a web application that provides easy access to millions of data points that help educators understand the performance of all students in a mid-western state.
- Worked with the team to build a microservice serving REST APIs using Spring Boot, delivering data from a data-warehouse in AWS Redshift, and a JS-based frontend to render the data as user-friendly reports.
- Backend - Java, Spring (Boot, JPA, Data, Data REST, MVC), JPA/Hibernate, Groovy/Spock, Microsoft SQL Server along with AWS Redshift
- Frontend - HTML/CSS, JSP, JavaScript (ES6, webpack, babel and jQuery), Jest.
- Infrastructure/Other - AWS (EC2, S3, RDS, ElasticBeanstalk), New Relic

EDUCATION

Master of Science, Computer Science

University of Minnesota

2013 - 2016